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REVIEW

The Circular Economy in Small and Medium-Sized Enterprises: A Systematic Review of Barriers and Drivers

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ABSTRACT

The transition to a circular economy based on the reuse, repair and recycling of resources requires the participation not only of large corporations but also of small and medium-sized enterprises, which account for a large share of employment and environmental impact. This systematic review synthesises empirical research on the barriers to and drivers of adopting circular practices in small and medium-sized businesses. A PRISMA-guided search identified 1,146 publications from 2015–2025, of which 72 studies from 28 countries met the inclusion criteria. Barriers and drivers were classified into five groups: financial, technological, market, institutional and organisational-cultural. The most frequently cited barriers were high upfront costs and limited access to finance (78% of studies), a lack of technical competence (61%) and low consumer demand for products made from secondary raw materials (47%). The key drivers are savings on raw material and disposal costs, requirements of large customers in supply chains and the personal values of owners. The role of government support is assessed ambiguously: subsidies are effective mainly in combination with advisory services. A substantial gap in research from Eastern Europe and Central Asia was identified. Recommendations for support policies and directions for further research are formulated.

Keywords: circular economy; small and medium-sized enterprises; systematic review; sustainable development; supply chains

Циркулярная экономика в малом и среднем бизнесе: систематический обзор барьеров и драйверов

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АННОТАЦИЯ

Переход к циркулярной экономике, основанной на повторном использовании, ремонте и переработке ресурсов, требует участия не только крупных корпораций, но и малых и средних предприятий, на долю которых приходится значительная часть занятости и воздействия на окружающую среду. В систематическом обзоре обобщены результаты эмпирических исследований барьеров и драйверов внедрения циркулярных практик в малом и среднем бизнесе. Поиск по протоколу PRISMA выявил 1 146 публикаций за 2015–2025 гг., из которых по критериям включения отобрано 72 исследования из 28 стран. Барьеры и драйверы классифицированы по пяти группам: финансовые, технологические, рыночные, институциональные и организационно-культурные. Наиболее часто упоминаемыми барьерами оказались высокие первоначальные затраты и ограниченный доступ к финансированию (78 % исследований), недостаток технических компетенций (61 %) и низкий спрос потребителей на продукцию из вторичного сырья (47 %). Ключевыми драйверами выступают экономия затрат на сырьё и утилизацию, требования крупных заказчиков в цепочках поставок и личные ценности собственников. Показано, что роль государственной поддержки оценивается неоднозначно: субсидии эффективны прежде всего в сочетании с консультационными услугами. Выявлен существенный дефицит исследований из стран Восточной Европы

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и Центральной Азии. Сформулированы рекомендации для политики поддержки и направления дальнейших исследований.
Ключевые слова: циркулярная экономика; малый и средний бизнес; систематический обзор; устойчивое развитие; цепочки поставок

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1 INTRODUCTION

The concept of the circular economy implies abandoning the linear “take–make–dispose” model in favour of closed loops in which products and materials retain their value for as long as possible. A considerable part of research and policy programmes is oriented towards large companies that have the resources to restructure their production processes.

Small and medium-sized enterprises face different constraints: they have less capital, fewer competencies and less bargaining power in supply chains. At the same time, they are more flexible and closer to local communities. The aim of this review is to systematise empirical evidence on the factors that hinder and facilitate the adoption of circular practices in small and medium-sized businesses.

2 MATERIALS AND METHODS

The review was conducted in accordance with the PRISMA 2020 guidelines. Bibliographic databases were searched using combinations of the terms “circular economy”, “small and medium-sized business” and their Russian-language equivalents. Empirical studies (surveys, case studies, interviews, econometric studies) published in 2015–2025 in English or Russian were included.

Of the 1,146 records retrieved, 214 studies underwent full-text assessment after removal of duplicates and screening of abstracts, and 72 met the inclusion criteria. Barriers and drivers were coded independently by two researchers according to a pre-developed scheme; inter-coder agreement was $\kappa = 0.79$.

3 RESULTS

The included studies cover 28 countries, but 61% of them are based on data from Western European countries. There are only five studies on the countries of Eastern Europe and Central Asia. The frequency with which the main barriers are mentioned is presented in Fig. 1.

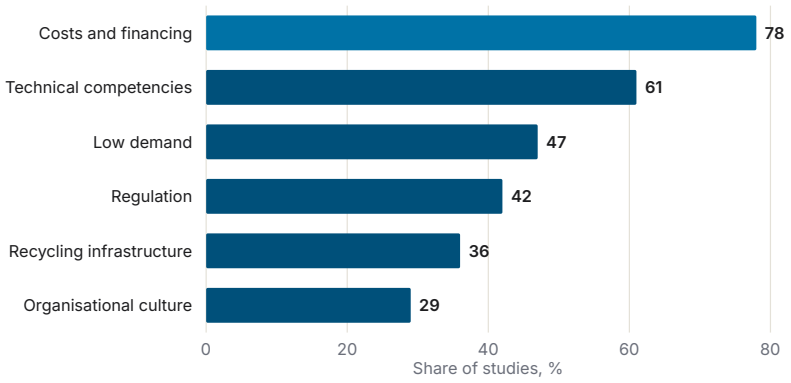


Fig. 1. Frequency of mention of barriers in the included studies, % (n = 72)

The drivers of the adoption of circular practices are grouped in Table 1. Economic motives — reducing the costs of raw materials and waste disposal — are mentioned most often, but in studies of family businesses the owners’ values play a substantial role.

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Table 1. Main drivers of the adoption of circular practices

Driver group	Example	Number of studies	Share, %
Financial	Savings on raw materials and disposal	51	71
Market	Requirements of large customers	38	53
Organisational and cultural	Owners' values	31	43
Institutional	Subsidies and advisory services	26	36
Technological	Digital exchange platforms	17	24

4 DISCUSSION

The dominance of financial barriers and drivers shows that for small businesses circularity is primarily a matter of economics rather than of the environmental agenda. Support measures that reduce initial costs and risks are therefore in greatest demand. At the same time, subsidies without advisory support often fail to bring about lasting change.

The geographical bias of the evidence base limits the transferability of the conclusions: the institutional environment, access to finance and recycling infrastructure differ substantially across countries. Filling this gap is an important task for regional research.

5 CONCLUSION

The adoption of circular practices in small and medium-sized businesses is determined primarily by economic incentives and supply chain requirements. Effective policy should combine financial support with the development of competencies and infrastructure.

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